

The Bischof–Stewart Pivoting Strategy: Mathematics, Complexity, and Pseudocode

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Abstract

We give a self-contained exposition of the column pivoting strategy based on incremental condition estimation, originally proposed by Bischof [2] and developed for column selection by Stewart [8]. The key idea is to choose pivots that greedily minimize the per-step growth of $\|R^{-1}\|_F$, where R is the upper-triangular factor of the evolving QR factorization. We derive the incremental update formulas, present detailed pseudocode using 1-indexed matrices, and analyze the computational complexity. Combined with Osinsky’s analysis [7], the resulting algorithm selects k columns from a $k \times n$ matrix with orthonormal rows in $O(nk^2)$ operations, achieving near-optimal submatrix conditioning guarantees.

1 Introduction

In the Golub–Klema–Stewart (GKS) algorithm [4] for column subset selection, one computes the leading k right singular vectors $V_k \in \mathbb{R}^{n \times k}$ of a matrix $A \in \mathbb{R}^{m \times n}$ and then selects k columns by applying a column-pivoted QR factorization to $V_k^T \in \mathbb{R}^{k \times n}$. Since V_k has orthonormal columns, V_k^T has orthonormal rows, and column selection reduces to finding a $k \times k$ submatrix $V_{J,1:k}$ with small $\|V_{J,1:k}^{-1}\|_F$ (or equivalently, large $\sigma_{\min}(V_{J,1:k})$).

The standard Businger–Golub [3] pivoting strategy selects the column of largest norm at each step. While efficient, it does not directly control the conditioning of the selected submatrix. Bischof [2] introduced *incremental condition estimation*, a technique that tracks the growth of $\|R^{-1}\|_F$ as columns are appended to the triangular factor during QR factorization. Stewart [8] adapted this idea to column selection: at each step, the pivot is chosen to minimize the incremental growth of $\|R^{-1}\|_F$.

Osinsky [7] proved that this greedy strategy, when applied to a matrix with orthonormal rows, achieves

$$\|V_{J,1:k}^{-1}\|_F \leq \sqrt{k(n-k+1)} \quad \text{and} \quad \|V_{J,1:k}^{-1}\|_2 \leq \sqrt{1+k(n-k)}, \quad (1)$$

matching the asymptotic guarantees of Gu–Eisenstat strong rank-revealing QR [5]. These bounds feed directly into the GKS error analysis: the spectral norm column selection error satisfies $\|A - A_{:,J}(A_{:,J})^\dagger A\|_2 \leq \sigma_{k+1}(A)\sqrt{1+k(n-k)}$, while the Frobenius norm error satisfies $\|A - A_{:,J}(A_{:,J})^\dagger A\|_F \leq \|\Sigma_\perp\|_F \sqrt{1+k(n-k)/r_k}$ [1].

The purpose of this note is to carefully derive the mathematics underlying the Bischof–Stewart criterion, present complete pseudocode with 1-indexed matrices, and analyze the computational cost.

2 Incremental Update of the Triangular Inverse

We work with a matrix $M \in \mathbb{R}^{k \times n}$ satisfying $MM^T = I_k$ (in the GKS context, $M = V_k^T$). A column-pivoted QR factorization builds, over k steps, a permutation Π and orthogonal Q such that

$$QM\Pi = \begin{pmatrix} R \\ 0 \end{pmatrix}$$

with $R \in \mathbb{R}^{k \times n}$ upper trapezoidal (here R is $k \times n$ since M already has k rows; the zero block is vacuous and Q is $k \times k$ orthogonal). More precisely, after k Householder steps one obtains $QM\Pi = R$ with R upper trapezoidal.

We describe the state after $i - 1$ completed steps ($1 \leq i \leq k$) and derive the formula governing the i -th pivot selection.

2.1 State after $i - 1$ steps

After $i - 1$ steps, orthogonal transformations Q_{i-1} have been applied from the left and columns have been permuted so that

$$Q_{i-1}^T M \Pi_{i-1} = \begin{pmatrix} R_{11} & R_{12} \\ 0 & \widetilde{M} \end{pmatrix}, \quad (2)$$

where $R_{11} \in \mathbb{R}^{(i-1) \times (i-1)}$ is upper triangular with nonzero diagonal, $R_{12} \in \mathbb{R}^{(i-1) \times (n-i+1)}$, and $\widetilde{M} \in \mathbb{R}^{(k-i+1) \times (n-i+1)}$ is the unreduced block. The first $i - 1$ columns (post-permutation) have been fully reduced; the remaining $n - i + 1$ columns are candidates for the i -th pivot.

2.2 Extending the factorization by one column

Suppose we select column j from the unreduced block (i.e., the j -th of the $n - i + 1$ remaining columns) as the next pivot. Denote by

$$r_j = R_{12}(:, j) \in \mathbb{R}^{i-1} \quad \text{and} \quad \tilde{m}_j = \widetilde{M}(:, j) \in \mathbb{R}^{k-i+1}$$

the top and bottom portions of this column in (2). A Householder reflection H_i is constructed to zero out the entries of $\tilde{m}_j$ below its first component, producing the new diagonal element

$$\rho_j = \|\tilde{m}_j\|_2. \quad (3)$$

After swapping column j into position i and applying H_i , the triangular factor grows from R_{11} to the $i \times i$ matrix

$$R_{11}^{(i)} = \begin{pmatrix} R_{11} & r_j \\ 0 & \rho_j \end{pmatrix}. \quad (4)$$

2.3 Incremental inverse formula

The inverse of the extended triangular factor (4) is

$$(R_{11}^{(i)})^{-1} = \begin{pmatrix} R_{11}^{-1} & -R_{11}^{-1}r_j/\rho_j \\ 0 & 1/\rho_j \end{pmatrix} = \begin{pmatrix} R_{11}^{-1} & -w_j/\rho_j \\ 0 & 1/\rho_j \end{pmatrix}, \quad (5)$$

where we define

$$w_j = R_{11}^{-1}r_j \in \mathbb{R}^{i-1}. \quad (6)$$

This is verified by direct multiplication: $R_{11}^{(i)} \cdot (R_{11}^{(i)})^{-1}$ yields the identity.

2.4 Growth of the inverse Frobenius norm

Taking the squared Frobenius norm of (5):

$$\begin{aligned}\|(R_{11}^{(i)})^{-1}\|_F^2 &= \|R_{11}^{-1}\|_F^2 + \frac{\|w_j\|_2^2}{\rho_j^2} + \frac{1}{\rho_j^2} \\ &= \|R_{11}^{-1}\|_F^2 + \frac{1 + \|w_j\|_2^2}{\rho_j^2}.\end{aligned}\tag{7}$$

The second term is the *increment* in $\|R^{-1}\|_F^2$ caused by appending column j .

2.5 The Bischof–Stewart pivot criterion

The Bischof–Stewart strategy [2, 8] selects the pivot that minimizes this increment. At step i , define the criterion for each remaining column $j \in \{i, i+1, \dots, n\}$ (after the permutation applied so far):

$$c_j = \frac{1 + \|w_j\|_2^2}{\rho_j^2} = \frac{1 + \|R_{11}^{-1} R_{12}(:, j)\|_2^2}{\|\widetilde{M}(:, j)\|_2^2}.\tag{8}$$

The pivot is

$$j^* = \underset{j \in \{i, \dots, n\}}{\operatorname{argmin}} c_j.\tag{9}$$

This greedy rule ensures that each step contributes as little as possible to $\|R^{-1}\|_F^2$.

Remark 2.1 (Interpretation). The numerator $1 + \|w_j\|_2^2$ measures how much the existing inverse “resists” the new column: if r_j is nearly in the column span of R_{11} , then $\|w_j\|_2$ is large and the criterion penalizes this choice. The denominator $\rho_j^2 = \|\tilde{m}_j\|_2^2$ rewards columns whose unreduced portion is large, as these produce a large diagonal element ρ_j and hence a small contribution $1/\rho_j^2$ to the inverse. The criterion balances these two effects.

Remark 2.2 (Relation to the spectral norm). While the criterion directly controls $\|R^{-1}\|_F$, the spectral norm $\|R^{-1}\|_2$ is also bounded. Since $\|R^{-1}\|_2 \leq \|R^{-1}\|_F$, any bound on the Frobenius norm of the inverse implies a (possibly looser) bound on the spectral norm. Osinsky [7] establishes the sharper spectral bound $\|R^{-1}\|_2 \leq \sqrt{1 + k(n-k)}$ directly.

2.6 Incremental maintenance of the w vectors

A naïve implementation would recompute $w_j = R_{11}^{-1} r_j$ from scratch at each step i for every remaining column j , at a cost of $O(i^2)$ per column via triangular solve. Over all steps and columns, this would total $O(nk^3)$ —unacceptably expensive. The key to an efficient implementation is that the w vectors can be updated incrementally.

After step i selects pivot j^* , the triangular factor grows from $R_{11} \in \mathbb{R}^{(i-1) \times (i-1)}$ to $R_{11}^{(i)} \in \mathbb{R}^{i \times i}$ as in (4). For a remaining column $\ell \neq j^*$, let $r_\ell^{(i)} \in \mathbb{R}^i$ denote its top i entries in the updated partially reduced matrix (after the Householder reflection H_i has been applied). Write

$$r_\ell^{(i)} = \begin{pmatrix} r_\ell \\ \alpha_\ell \end{pmatrix}$$

where $r_\ell \in \mathbb{R}^{i-1}$ is the portion from the previous step and α_ℓ is the new i -th entry produced by H_i . Then

$$w_\ell^{(i)} = (R_{11}^{(i)})^{-1} r_\ell^{(i)} = \begin{pmatrix} R_{11}^{-1} & -w_{j^*}/\rho_{j^*} \\ 0 & 1/\rho_{j^*} \end{pmatrix} \begin{pmatrix} r_\ell \\ \alpha_\ell \end{pmatrix} = \begin{pmatrix} w_\ell - (\alpha_\ell/\rho_{j^*}) w_{j^*} \\ \alpha_\ell/\rho_{j^*} \end{pmatrix},\tag{10}$$

where $w_\ell = R_{11}^{-1}r_\ell$ is the w -vector from the previous step. The update (10) requires only:

1. the scalar α_ℓ/ρ_{j^*} (one division, already available from the Householder step), and
2. a scaled vector subtraction $w_\ell \leftarrow w_\ell - (\alpha_\ell/\rho_{j^*})w_{j^*}$, costing $O(i)$ operations.

Across all $n - i$ remaining columns, the update costs $O(i(n - i))$ at step i , and summing over $i = 1, \dots, k$ gives a total of $O(nk^2)$ for maintaining all w -vectors.

Similarly, $\|w_\ell^{(i)}\|_2^2$ can be maintained incrementally:

$$\|w_\ell^{(i)}\|_2^2 = \|w_\ell\|_2^2 - 2\frac{\alpha_\ell}{\rho_{j^*}}w_\ell^T w_{j^*} + \frac{\alpha_\ell^2}{\rho_{j^*}^2}(\|w_{j^*}\|_2^2 + 1), \quad (11)$$

which costs $O(i)$ per column (dominated by the inner product $w_\ell^T w_{j^*}$, but this can be computed simultaneously with the vector update).

3 Algorithm

Algorithm 3 presents the complete Bischof–Stewart column-pivoted QR factorization. All indices are 1-based. The algorithm operates in-place on the input matrix M and returns the permutation vector π identifying the selected columns.

Remark 3.1 (Column norm downdating). Line 19 uses the standard norm downdating identity $\|M(i+1 : k, j)\|_2^2 = \|M(i : k, j)\|_2^2 - M(i, j)^2$. In floating-point arithmetic this subtraction can suffer cancellation when $M(i, j)^2 \approx \|M(i : k, j)\|_2^2$. The classical remedy, as in the Businger–Golub algorithm [3], is to recompute the column norm from scratch whenever the downdated value drops below a threshold (e.g., when more than half the original norm has been subtracted). This safeguard adds at most $O(nk)$ operations over the entire algorithm and does not change the asymptotic cost.

Remark 3.2 (First step simplification). At $i = 1$, the triangular factor R_{11} is empty and w_j is vacuous, so the criterion reduces to $c(j) = 1/s(j)$. Minimizing $c(j)$ is equivalent to maximizing $s(j) = \|M(:, j)\|_2^2$, which coincides with the Businger–Golub rule. Thus the first pivot of the Bischof–Stewart strategy agrees with standard column-pivoted QR.

Remark 3.3 (Tracking the full inverse norm). If the running value of $\|R^{-1}\|_F^2$ is desired (e.g., for diagnostic purposes), it can be maintained by initializing $F \leftarrow 0$ and updating $F \leftarrow F + c(j^*)$ after each pivot selection. After all k steps, $F = \|R^{-1}\|_F^2$ where $R = R_{11}^{(k)}$ is the final $k \times k$ triangular factor.

4 Computational Complexity

We analyze the cost of Algorithm 3 step by step. Throughout, we assume $k \leq n$ and focus on the regime $k \ll n$ relevant to the GKS application.

4.1 Per-step costs

At step i (selecting the i -th pivot), the dominant operations are:

1. **Criterion evaluation** (lines 6–10): Computing $c(j)$ for each of the $n - i + 1$ remaining columns requires reading $\|W(1 : i - 1, j)\|_2^2$ and $s(j)$. If $\|W(1 : i - 1, j)\|_2^2$ is maintained as a running scalar (updated alongside W via (11)), each evaluation costs $O(1)$, for a total of $O(n - i)$ at step i .

Algorithm 1: Bischof–Stewart Column-Pivoted QR**Input:** $M \in \mathbb{R}^{k \times n}$ with $MM^T = I_k$, $k \leq n$ **Output:** Permutation π such that $\sigma_{\min}(M(:, \pi(1:k)))$ is controlled

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1:  $\pi \leftarrow (1, 2, \dots, n)$  % initialize permutation
2:  $W \leftarrow$  empty array of size  $0 \times n$  %  $W(:, j)$  stores  $R_{11}^{-1} R_{12}(:, j)$ 
3:  $s(j) \leftarrow \|M(:, j)\|_2^2$  for  $j = 1, \dots, n$  % squared column norms
4: for  $i = 1, \dots, k$  do
   % — Pivot selection —
5:   for  $j = i, \dots, n$  do
6:     if  $i = 1$ :  $c(j) \leftarrow 1/s(j)$  %  $w_j$  is empty
7:     else:  $c(j) \leftarrow (1 + \|W(1:i-1, j)\|_2^2)/s(j)$ 
8:    $j^* \leftarrow \operatorname{argmin}_{j \in \{i, \dots, n\}} c(j)$ 
   % — Swap columns  $i$  and  $j^*$  —
9:   Swap  $\pi(i) \leftrightarrow \pi(j^*)$ ; swap  $M(:, i) \leftrightarrow M(:, j^*)$ 
10:  Swap  $W(:, i) \leftrightarrow W(:, j^*)$ ; swap  $s(i) \leftrightarrow s(j^*)$ 
   % — Householder reduction of column  $i$  —
11:  Compute Householder  $H_i$  s.t.  $H_i M(i:k, i) = \rho_i e_1$ ,  $\rho_i = \|M(i:k, i)\|_2$ 
12:   $M(i:k, i:n) \leftarrow H_i M(i:k, i:n)$ 
   % — Update  $W$  and  $s$  for remaining columns —
13:  if  $i < k$  then
14:     $w^* \leftarrow W(1:i-1, i)$  %  $w$ -vector of chosen pivot
15:    for  $j = i+1, \dots, n$  do
16:       $\alpha_j \leftarrow M(i, j)$ ;  $\beta_j \leftarrow \alpha_j/\rho_i$ 
17:      if  $i = 1$ :  $W(1, j) \leftarrow \beta_j$ 
18:      else:  $W(1:i-1, j) \leftarrow W(1:i-1, j) - \beta_j \cdot w^*$ ;  $W(i, j) \leftarrow \beta_j$ 
19:       $s(j) \leftarrow s(j) - \alpha_j^2$  % downdate column norm
20:  return  $\pi$ 

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2. **Column swaps** (lines 13–16): Swapping columns in M , W , s , and π costs $O(k)$ for M , $O(i)$ for W , and $O(1)$ for s and π ; total $O(k)$.
3. **Householder reflection** (lines 18–19): Constructing H_i costs $O(k-i)$. Applying H_i to the submatrix $M(i:k, i:n)$ of size $(k-i+1) \times (n-i+1)$ costs $O((k-i)(n-i))$.
4. **W -vector update** (lines 22–30): For each of the $n-i$ remaining columns, the update (10) costs $O(i)$ (a scaled vector subtraction of length $i-1$ plus one scalar division). Total: $O(i(n-i))$.
5. **Norm downdating** (line 31): One subtraction per remaining column: $O(n-i)$.

4.2 Total cost

Summing over $i = 1, \dots, k$:

Householder reflections.

$$\sum_{i=1}^k O((k-i)(n-i)) = O\left(n \sum_{i=1}^k (k-i)\right) = O\left(n \cdot \frac{k(k-1)}{2}\right) = O(nk^2).$$

W -vector maintenance.

$$\sum_{i=1}^k O(i(n-i)) = O\left(n \sum_{i=1}^k i - \sum_{i=1}^k i^2\right) = O\left(n \cdot \frac{k(k+1)}{2}\right) = O(nk^2).$$

Criterion evaluation and norm downdating.

$$\sum_{i=1}^k O(n-i) = O(nk).$$

Column swaps.

$$\sum_{i=1}^k O(k) = O(k^2).$$

The total cost is therefore

$$\boxed{T(n, k) = O(nk^2)} \tag{12}$$

operations, dominated equally by the Householder reflections and the W -vector updates.

4.3 Comparison with alternative strategies

1. **Businger–Golub column-pivoted QR** [3]: Also $O(nk^2)$, but uses maximum-column-norm pivoting, which does not directly control $\|R^{-1}\|_F$ and hence provides weaker conditioning guarantees.
2. **Gu–Eisenstat strong RRQR** [5]: Achieves optimal bounds on the entries of $R_{11}^{-1}R_{12}$, but the algorithm involves a post-processing phase of column swaps. Each swap requires re-triangularizing via Givens rotations, and the number of swaps is bounded but can be substantial. The worst-case cost exceeds $O(nk^2)$ when many swaps are needed. In contrast, Bischof–Stewart achieves comparable asymptotic guarantees in a single forward pass.
3. **Standard partial QR (no pivoting)**: $O(nk^2)$, but without any pivoting the selected submatrix may be arbitrarily ill-conditioned.

Remark 4.1 (Practical considerations). The W -vector updates involve BLAS-1 operations (scaled vector addition) that are memory-bandwidth limited on modern hardware. In practice, the updates can be organized as a matrix operation: defining $W_{\text{rem}} \in \mathbb{R}^{(i-1) \times (n-i)}$ as the block of w -vectors for remaining columns, the update at step i is a rank-1 modification $W_{\text{rem}} \leftarrow W_{\text{rem}} - (\beta/\rho_i) w^* \cdot \alpha^T$, where $\alpha \in \mathbb{R}^{n-i}$ collects the new row entries. This enables the use of BLAS-2 routines, improving cache utilization.

5 Connection to GKS Error Bounds

The output of Algorithm 3 is a permutation π whose first k entries identify the selected column indices $J = \{\pi(1), \dots, \pi(k)\}$. By Osinsky’s analysis [7], the selected submatrix $V_{J,1:k}$ satisfies the bounds (1).

These bounds enter the GKS error analysis through the master bound: for any orthonormal $Z \in \mathbb{R}^{n \times k}$ approximating the leading right singular subspace of A and any index set J with $|J| = k$,

$$\|A - A_{:,J}(A_{:,J})^\dagger A\|_2 \leq \frac{\|A - AZZ^T\|_2}{\sigma_{\min}(Z_J)}.$$

Setting $Z = V_k$ (exact singular vectors) and applying the Bischof–Stewart selection yields $\sigma_{\min}(V_{J,1:k}) \geq 1/\sqrt{1 + k(n - k)}$, giving the spectral bound $\|A - A_{:,J}(A_{:,J})^\dagger A\|_2 \leq \sigma_{k+1}(A)\sqrt{1 + k(n - k)}$. The Frobenius bound follows similarly from $\|V_{J,1:k}^{-1}\|_F^2 \leq k(n - k + 1)$ [1].

For the randomized GKS algorithm, the same column selection procedure is applied to approximate right singular vectors $\hat{V}_k$ obtained via randomized SVD. Since $\hat{V}_k$ also has orthonormal columns, Algorithm 3 applies with identical guarantees on $\sigma_{\min}(\hat{V}_{k,J})$, and the overall error bound involves an additional factor from the subspace approximation quality [6].

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